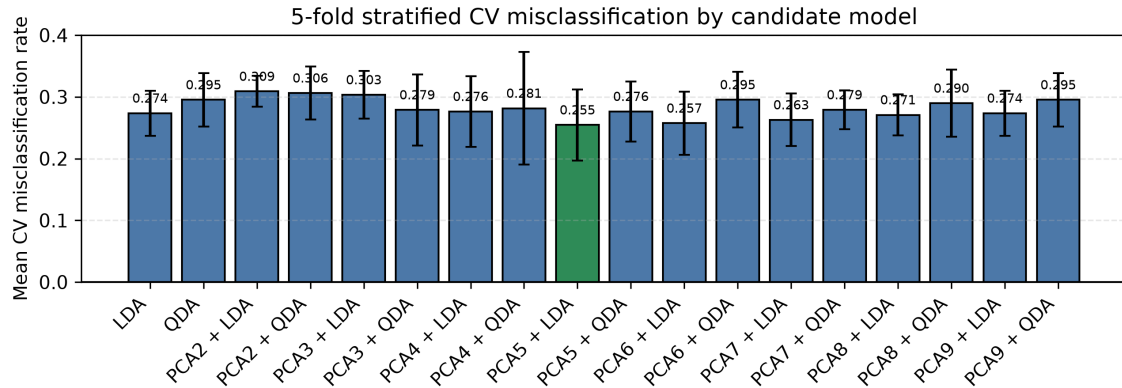


LDA/QDA analysis for the SAheart dataset

The binary response was `chd`; predictors were `sbp`, `tobacco`, `ldl`, `adiposity`, `famhist`, `typea`, `obesity`, `alcohol`, `age`, with `famhist` coded as **Absent** = 1 and **Present** = 2. A stratified 20% test split was reserved before selection using seed 432, leaving 369 training observations and 93 test observations. All 18 candidate LDA/QDA models were evaluated with the same 5-fold stratified CV assignments on the training data; preprocessing was fit only on each training fold.



CV comparison

All 18 candidates were evaluated; the table below reports the top models and the two ordinary baselines.

Candidate	Mean CV error	SD
LDA	0.2736	0.0364
QDA	0.2952	0.0431
PCA5 + LDA	0.2545	0.0576
PCA6 + LDA	0.2573	0.0509
PCA7 + LDA	0.2628	0.0426
PCA3 + QDA	0.2789	0.0577
PCA4 + LDA	0.2762	0.0569
PCA5 + QDA	0.2763	0.0486

Selected model and test performance

The selected model was PCA5 + LDA, with the lowest mean 5-fold CV misclassification rate of 0.2545. The final held-out test result for this model was 23/93 misclassified, corresponding to a test misclassification rate of 0.2473.

Confusion matrix on the reserved test set (rows = observed, columns = predicted):

Obs	Pred	0	1
0		52	9
1		14	18

Findings and limitations

PCA5 + LDA delivered the best cross-validated predictive performance among the considered LDA/QDA-based candidates, slightly improving over ordinary LDA. This suggests a modest gain from dimensionality reduction before classification, although the improvement is small relative to the overall classification difficulty.

The main limitations are the single 80/20 random split, the modest sample size, and the fact that preprocessing and model selection were based only on the training set. The held-out score should therefore be interpreted as an estimate rather than a guarantee of future performance.